



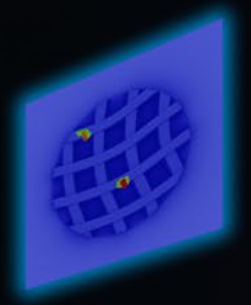
Vision System with Built-in AI

NEW VS Series



New AI Technology | **Reliable Defect Identification**

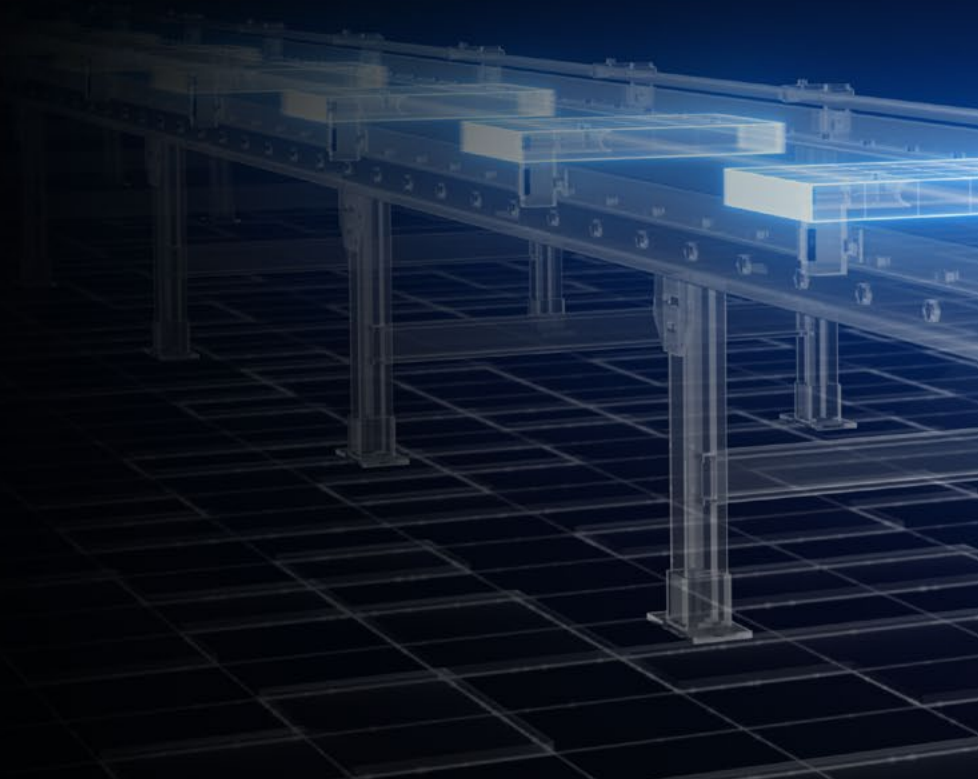
VS Series

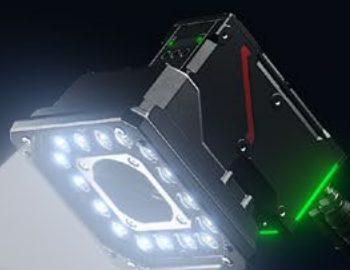
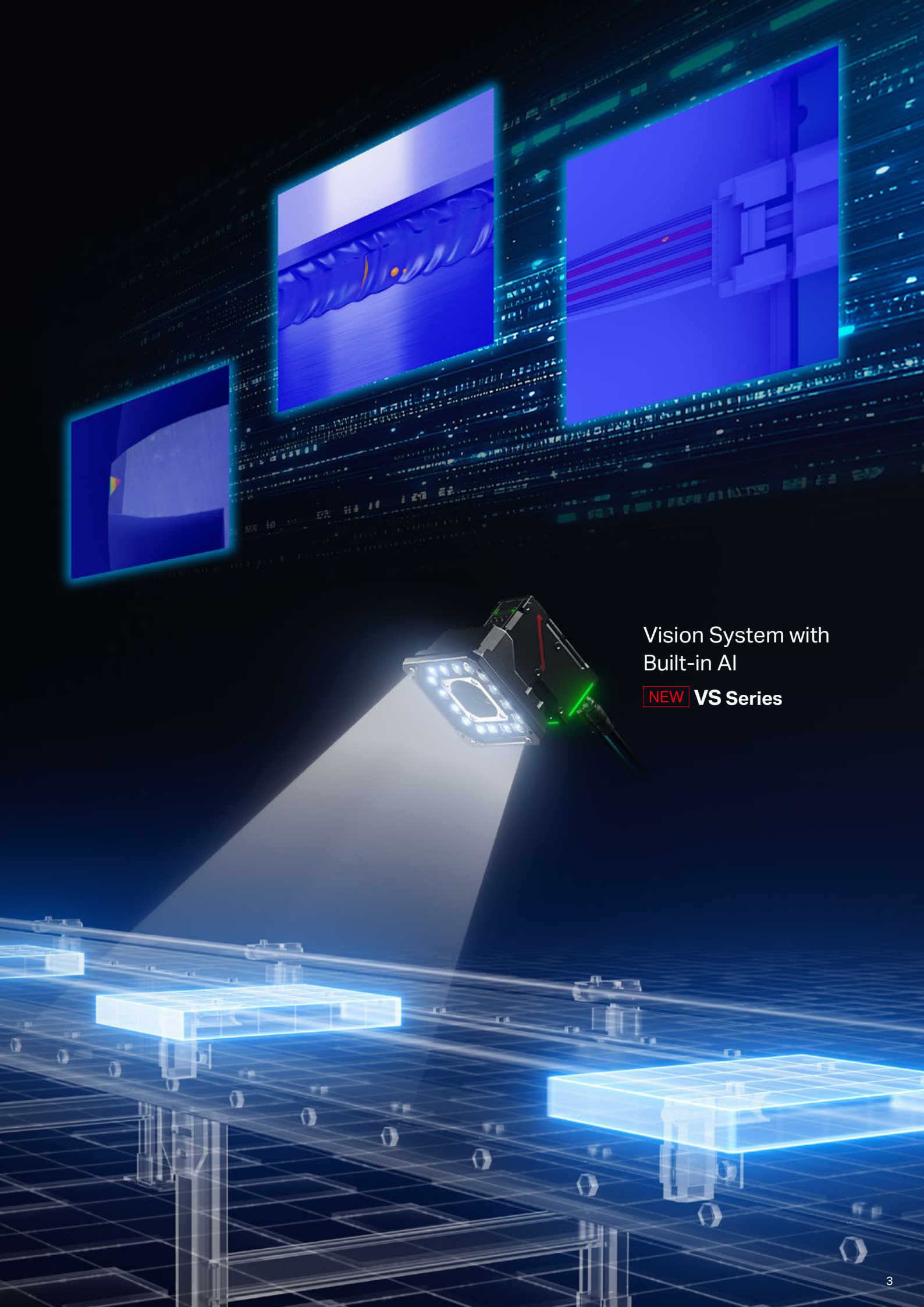


Two AI technologies for automation of previously impossible vision inspections

[Technology 1] Increased detection performance: Proprietary KEYENCE algorithm for smart camera-based 25-megapixel high-accuracy AI inspection

[Technology 2] Increased operational efficiency: Quick and easy startup using a general-purpose PC and only a few dozen images for learning





Vision System with
Built-in AI

NEW VS Series

Optimal tools × Optimal image creation to meet a wide variety of inspection needs



Lineup



Standard zoom
smart camera
VS-L



Short-range zoom
smart camera
VS-S



C-mount smart
camera
VS-C

Optimal inspection tools

[Two AI tools + Rule-based tools]

The combination of both supervised and unsupervised AI tools alongside rule-based inspection tools provide flexible solutions to a wide variety of vision applications.

Supervised
AI learning



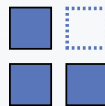
AI Segmentation

Unsupervised
AI learning

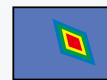


AI Detection

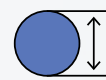
Rule-based tools



Presence
check



Scratches



GD&T
measurement

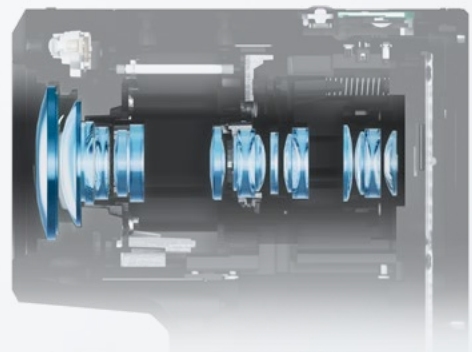


Count

Optimal image creation

[ZoomTrax × Specialised imaging]

Award-winning imaging and illumination tools such as LumiTrax, Multi-Spectrum, and ZoomTrax make it possible to create the ideal image for stable inspection.



Multi-spectrum lighting

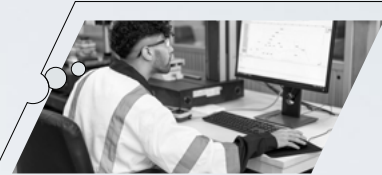


High-intensity smart ring
illumination

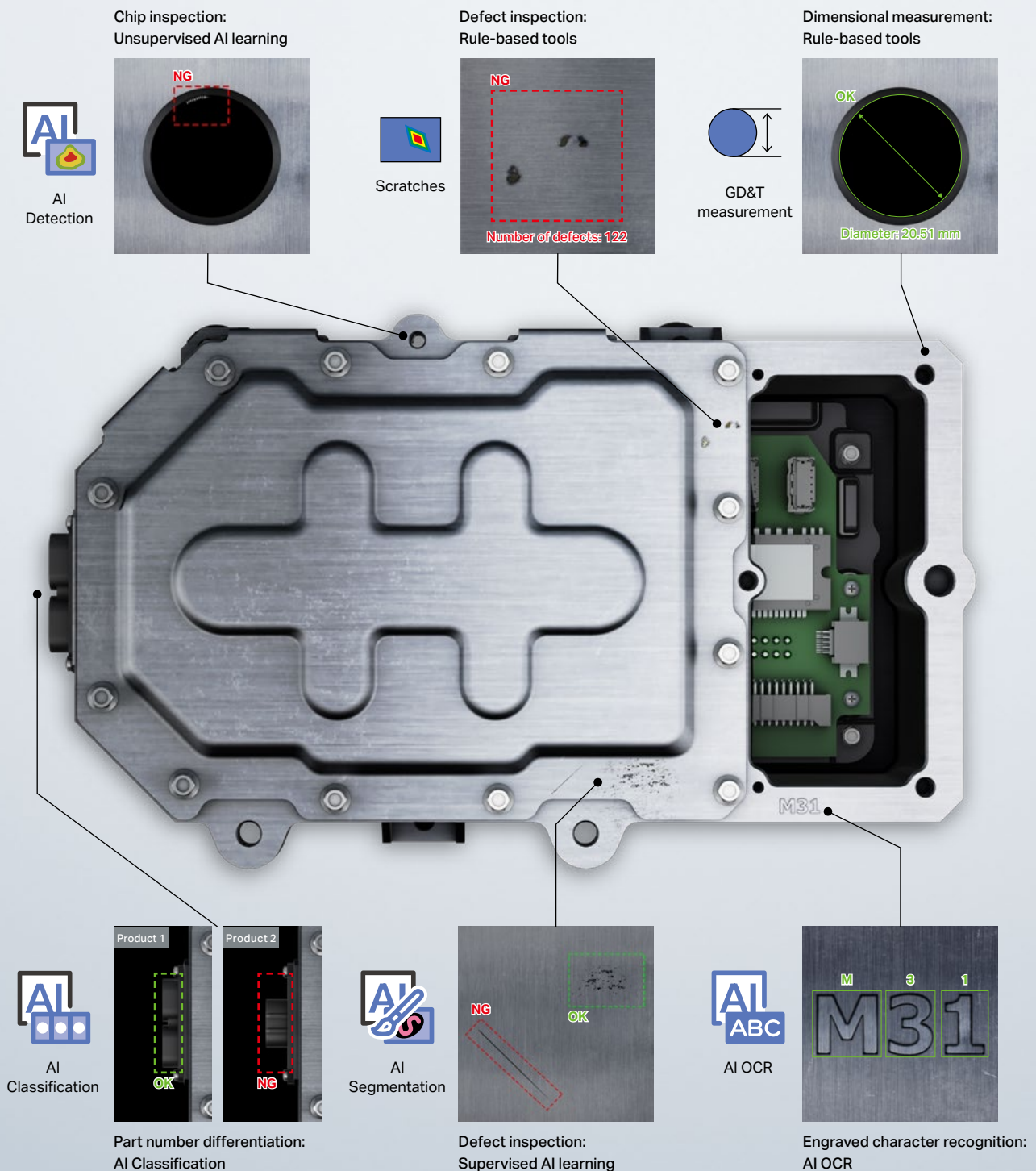
Optimal inspection tools

Two AI tools × Rule-based tools

Improving yield rates by reducing excessive detection of good products and minimising accidental shipment of defective products is incredibly difficult.



Improve flexibility through supervised and unsupervised AI learning + rule-based inspection tools



Application-specific AI tools × Rule-based tools

AI Tools



NEW Supervised AI learning
AI Segmentation



Unsupervised AI learning
AI Detection



Specialised for differentiation
AI Classification



NEW Specialised for OCR
AI OCR

Rule-based tools



Presence
check



Scratches



GD&T
measurement



Positioning



Colour
inspection



Count



Character
recognition



1D code



2D code

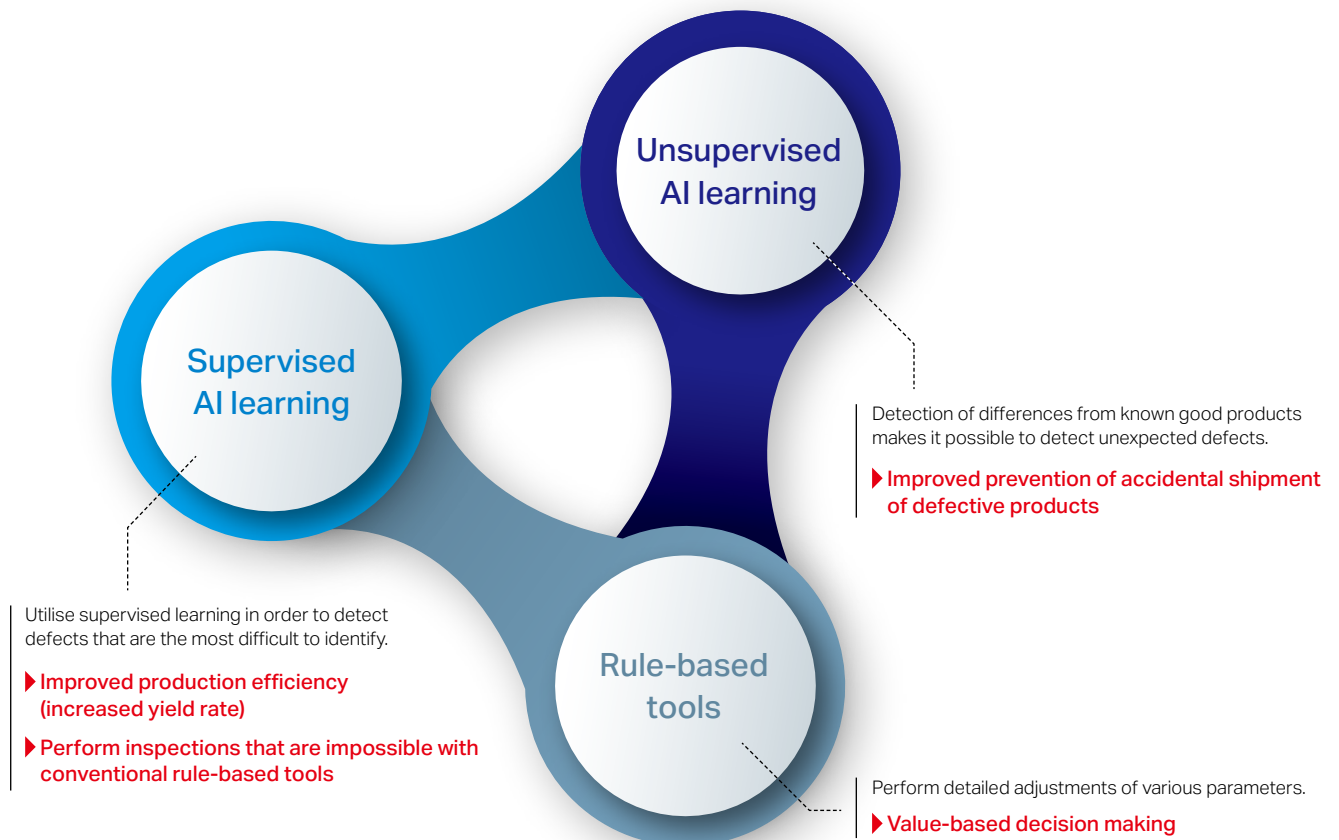


Calibration

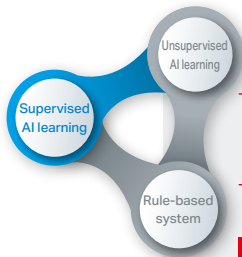
* The VS-AD1 is required to use AI Segmentation and AI Detection Fine Mode.

The right tool for the right application

Take advantage of two AI tools along with rule-based tools to supercharge any inspection.



Two AI tools × Rule-based tools



1 | Supervised AI Inspection NEW AI Segmentation

NG Scratches



Challenge with conventional appearance inspection

Excessive detection of defects

Varying part conditions lead to false detection of defects.



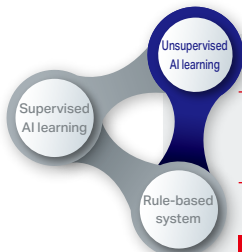
Solution with the VS Series

OK Stains



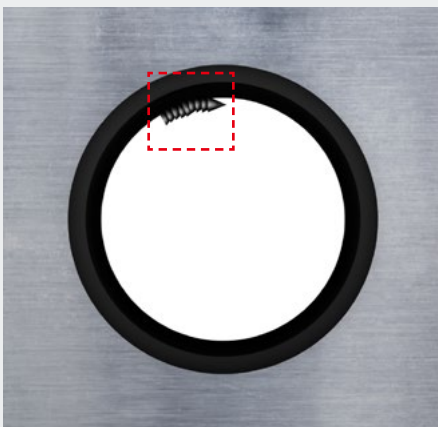
Detection of only specified defects

AI Segmentation makes it possible to learn specific defects, reducing excessive detection.



2 | Unsupervised AI Inspection AI Detection

NG Unexpected chips



Challenge with conventional appearance inspection

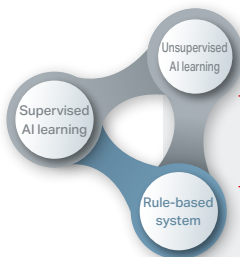
Unexpected defects may not be detected



Solution with the VS Series

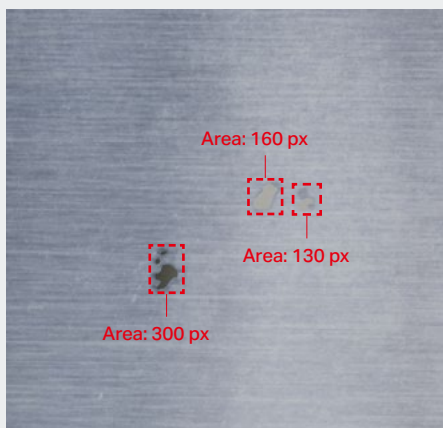
Detect defects by comparing products to the learned good conditions

Even unexpected defects can be detected by teaching only OK images.



3 Rule-Based Inspection

Detection based on size and count:
Cast holes



Challenge with conventional appearance inspection

AI results do not provide size or count results

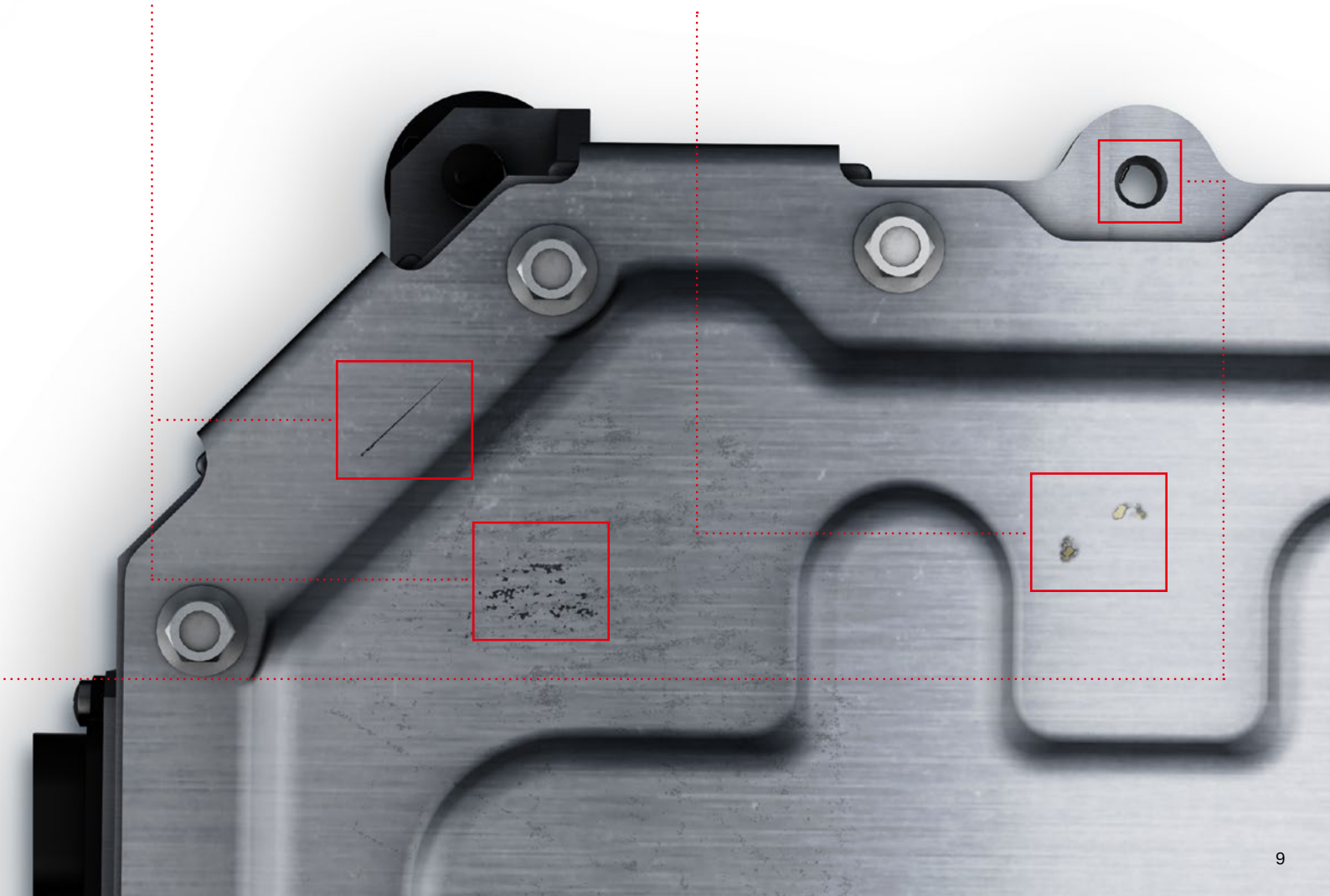
Setting judgement conditions for size or count are not possible with AI.



Solution with the VS Series

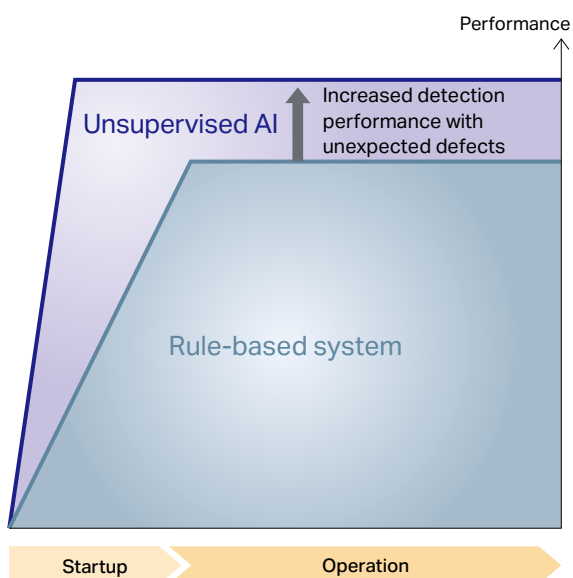
Detect size and count of defects that AI Tools flag using rule-based tools

Defective areas detected by AI tools can be re-inspected with rule-based inspection tools to set judgement conditions for defect size and count.



Three tools for optimising inspection performance

Prevention of accidental shipment of defective products



Conventional products

Rule-based system

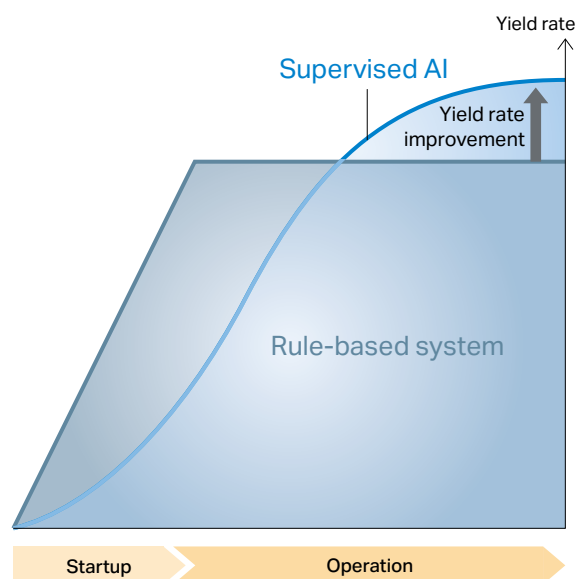
Startup time can be reduced and detection of even minute defects is possible, but additional setting configuration is needed to detect unexpected defects.

VS Series

Rule-based system × Unsupervised AI learning

The adoption of AI capable of detecting unexpected defects makes it possible to prevent accidental shipment of defective products.

Yield rate improvement



Conventional products

Rule-based system

Although startup time is short, detecting defects in a wide variety of good products increases excessive detection and decreases the yield rate.

VS Series

Rule-based system × Supervised AI learning

AI that can handle variations in good products without sacrificing operating speed makes it possible to improve yield rates.

Two AI tools × Rule-based system

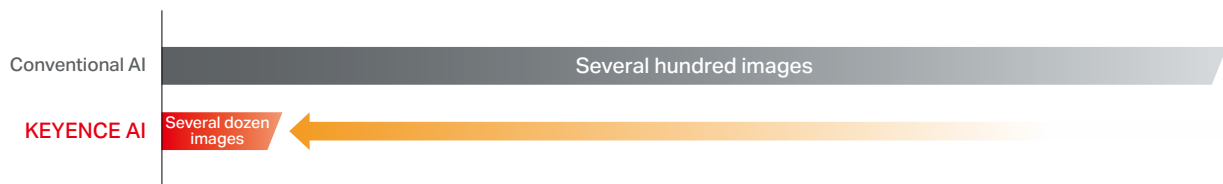


Simultaneous elimination of accidental shipment of defective products and improved yield rate

KEYENCE proprietary AI: Immediate usability with only a few sample products

Conventional AI learning requires a large number of images (hundreds to thousands) to achieve the desired detection performance. With the VS Series, however, the AI tools offer impressive detection performance with only a small number of images (from just a few to several dozen), significantly reducing startup time.

Number of images required for learning before operation



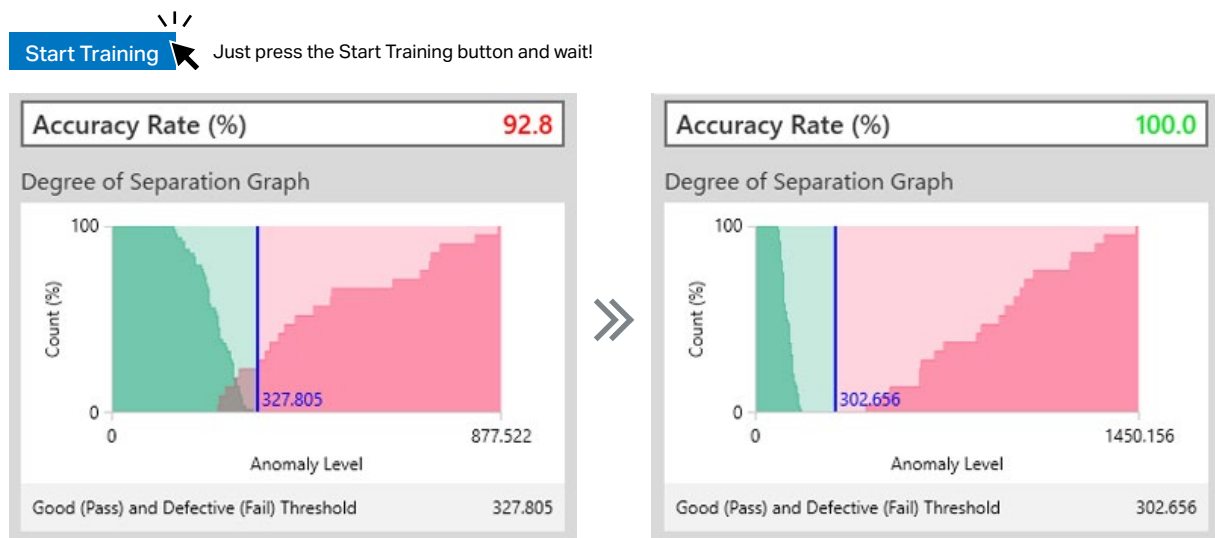
Maximum detection capability by learning just a small number of images



Significantly reduced startup time

Automatic selection of images for learning AI Auto Image Selector

With conventional AI systems, users have to select the images to use for learning manually, requiring a certain level of experience and a lot of time. With AI Auto Image Selector, the software automatically selects the images for learning, eliminating the need for specialised skills and significantly reducing the time needed for learning.



The Auto Image Selector automatically repeats learning until clear separation of results is possible.

No need for trial and error or special skills

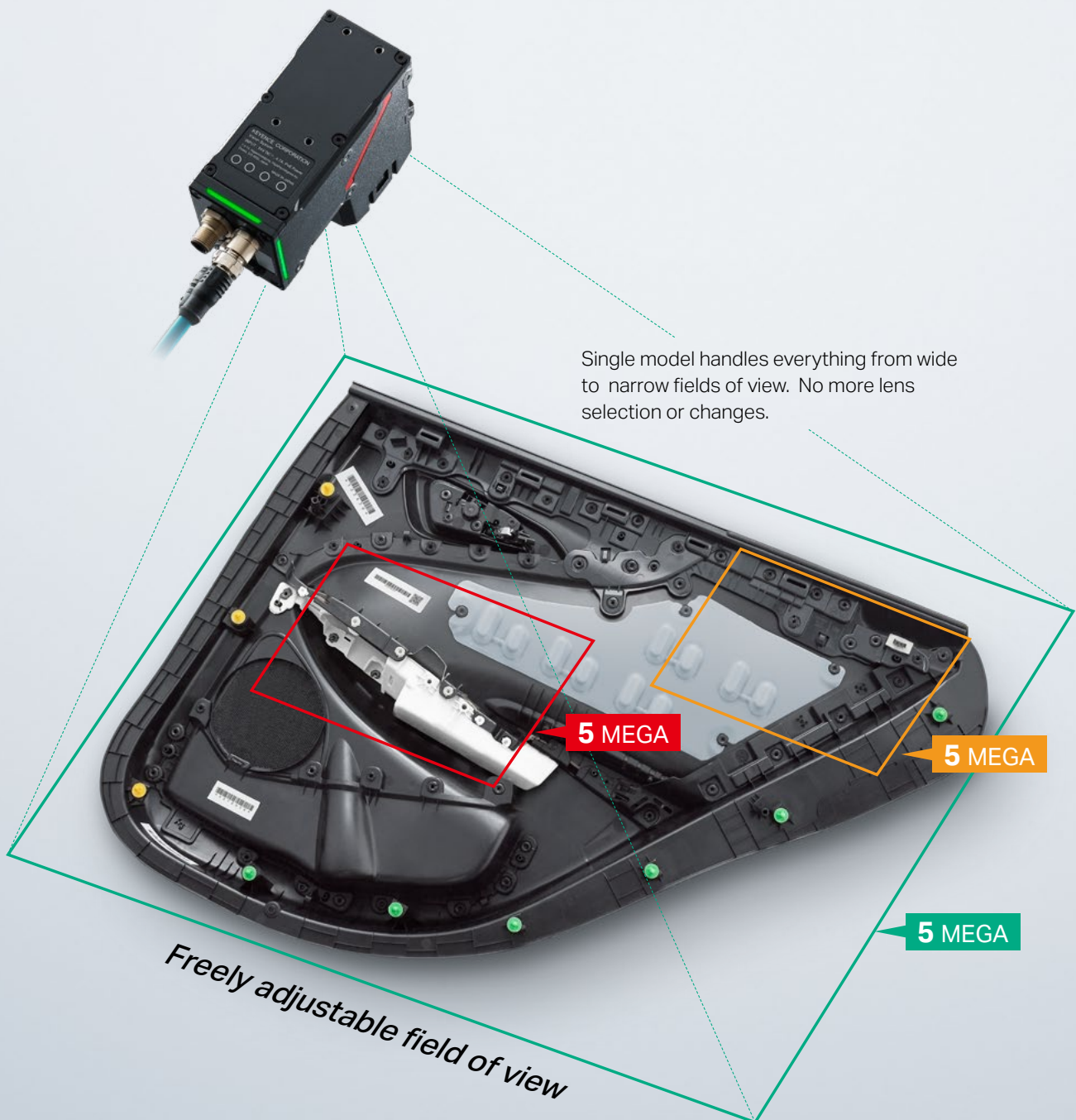


Significantly reduced time and effort spent on learning

Image optimisation used to require a great amount of time and skill.



Adjust focus and field of view with one click



Single model handles everything from wide to narrow fields of view. No more lens selection or changes.

5 MEGA

5 MEGA

5 MEGA

Freely adjustable field of view



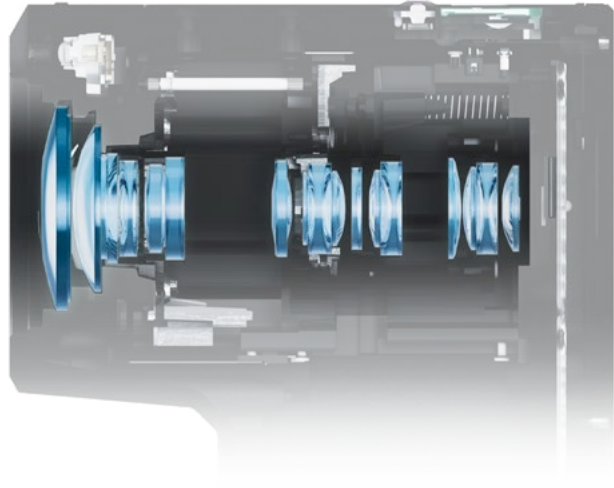
See the video
for more
information



Optical zoom with no loss in resolution: ZoomTrax

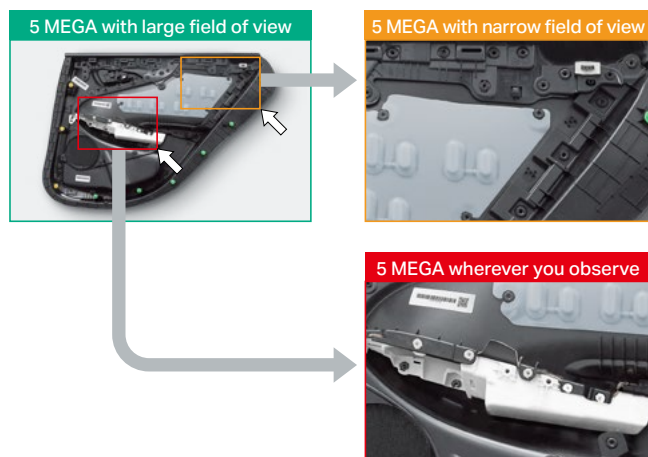
A series of mechanical zoom lenses allows a single camera to be used across a wide range of mounting distances and fields of view without changing lenses.

Customised CMOS sensors are available from 1.6 to 15 MP. By integrating a series of 19 lenses and a CMOS sensor, the VS Series realises high-grade optical performance with minimal distortion.



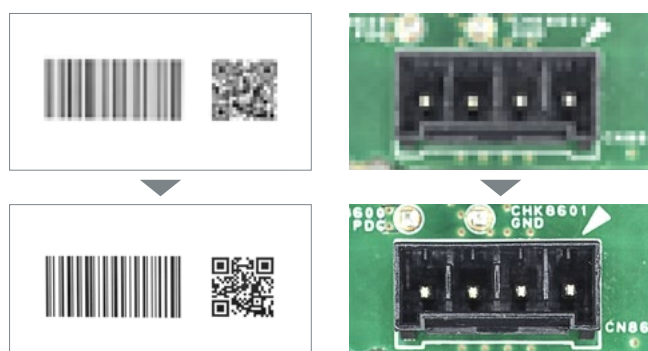
Simply specify the area to get the best field of view and resolution

Using optical zoom, the VS can easily change the field of view without sacrificing resolution. All adjustments can be completed from the software. No mechanical changes necessary!



Optimised images for robust inspections

The VS Series utilises a unique auto-focus algorithm to create the best image. This optimises images for code reading, measurement, and flaw detection applications that require high resolution.



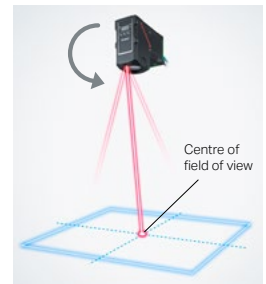
Our commitment to easy installation

We considered every possible issue that you may encounter with camera installation.
No control panel is required, so it can be installed immediately, anytime, before or during projects.

OLED Display

View connection status,
IP address, and other settings
directly on the camera.

Laser pointer for easy installation



Flexible mounting options

Ability to install on existing equipment by
utilising mounting holes on 4 sides.

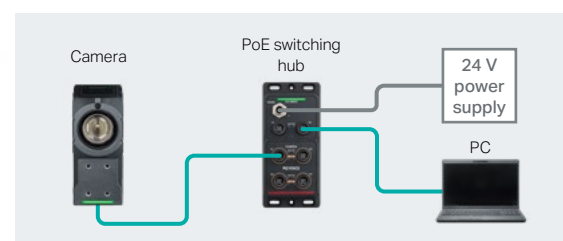
IP67 Rated

A double-packed structure ensures a high level of
waterproofing.



PoE

Supply power to both the camera and light with one cable
using Power over Ethernet.



Lighting techniques and algorithms for reliable inspections

KEYENCE's high-intensity smart ring light features snap-on installation and the ability to cut ambient light as well as advanced algorithms that adapt to changes in the environment.



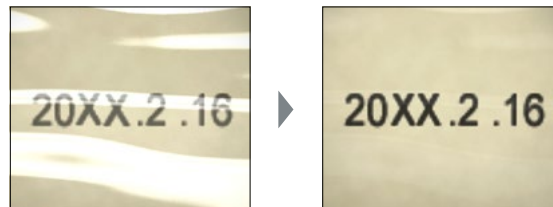
Fine HDR

Uniform images can be captured by correcting the contrast with one image capture. Multiple image captures are not required, so the VS Series can be used on high-speed lines.



LumiTrax Glare Reduction

LumiTrax mode combines smart ring lighting techniques with proprietary algorithms to minimise glare.

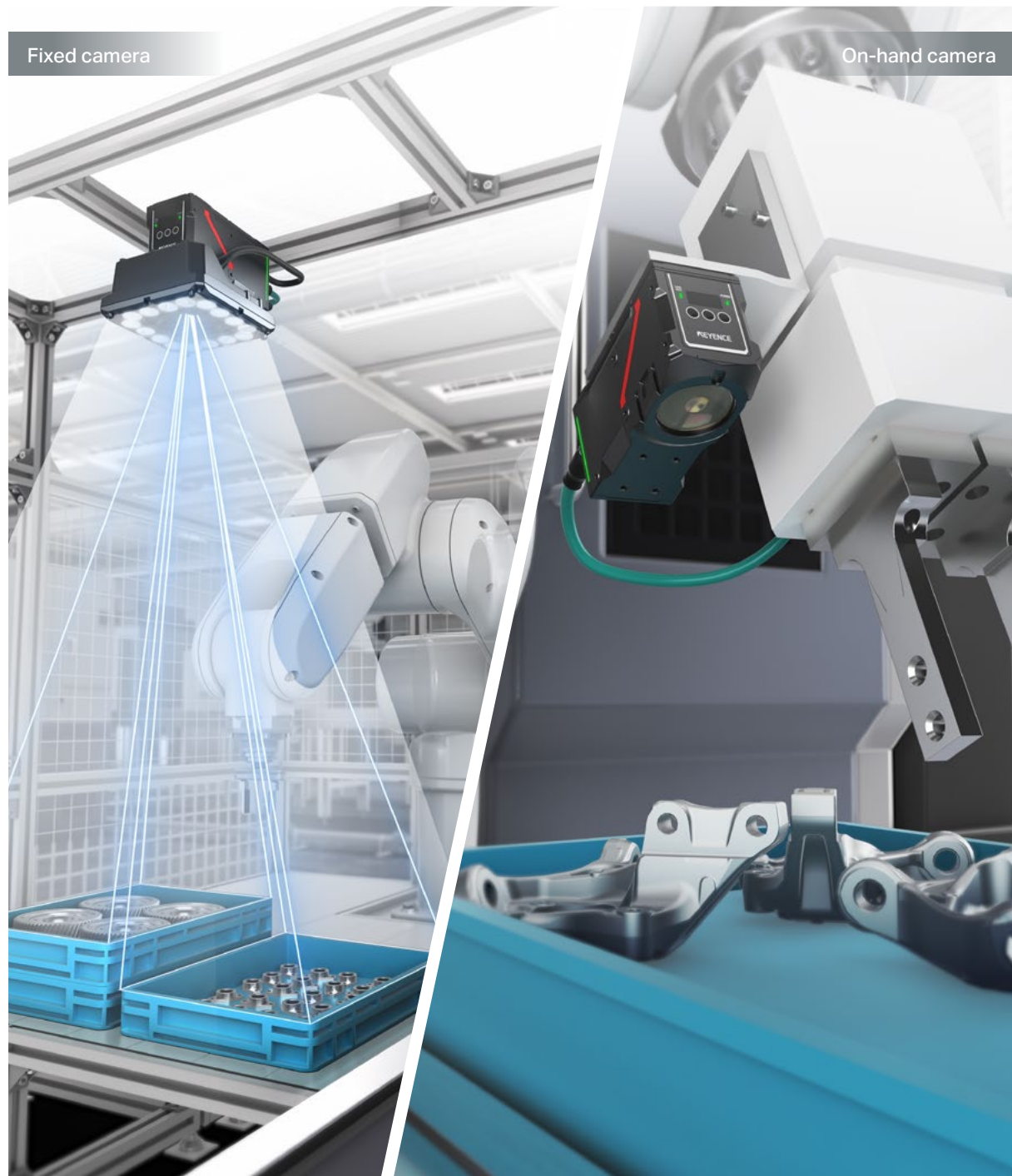


Optimal image creation with vision-guided robotics

Fixed and on-hand camera support

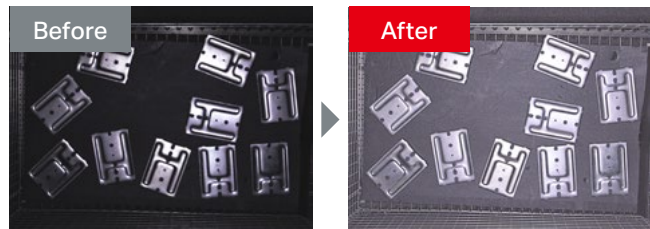
Improved stability of vision-guided robotics

The VS series improves vision-guided robotics applications by utilising ZoomTrax, smart cameras, and high intensity smart ring illumination functions for greater imaging and environmental stability.



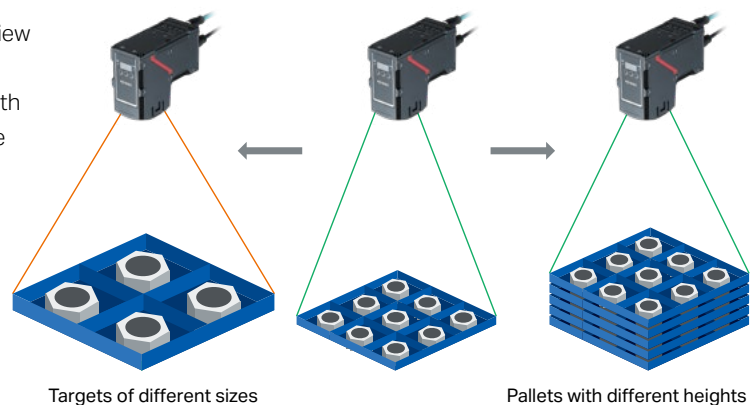
Bright, uniform illumination even over a wide field of view Fine HDR

With a wider field of view, brightness can be uneven with a noticeable difference in intensity between the centre and the edges of the screen. Advanced imaging technologies such as overdrive lighting with high-intensity smart rings and HDR capturing help to minimise such differences even with a wide field of view.



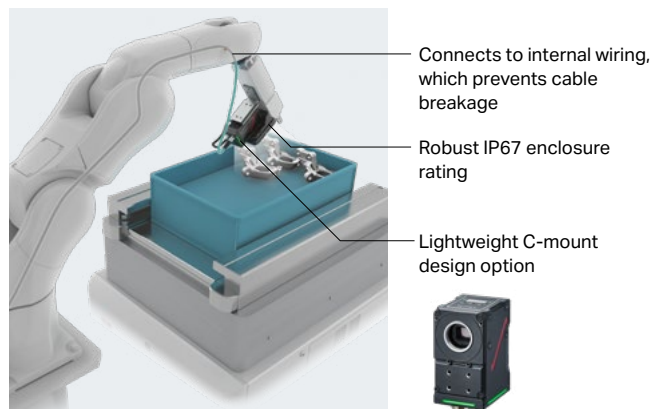
Support for multiple pallets and tooling changeover ZoomTrax

ZoomTrax automatically changes the field of view to suit the size of the target, and automatic focusing makes it possible to handle pallets with different heights, allowing for usability in a wide variety of production environments without changing lenses.



Optimal hardware for on-hand picking Internal wiring

A wider range of on-hand applications is possible thanks to on-robot PoE connectivity, IP67 environmental resistance, and a lightweight C-mount design.



VS Creator for quick programming of any inspection

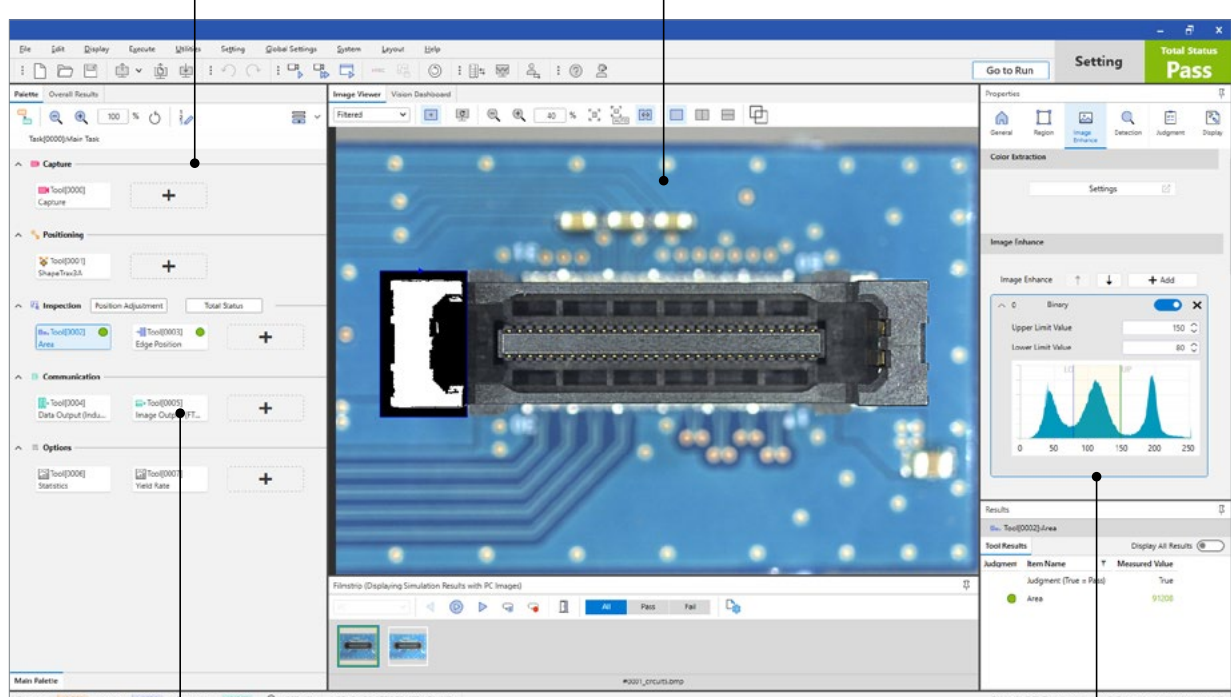
Create both simple and sophisticated inspections

Quick inspection creation with one software package

VS Creator is designed to be both easy to use and intuitive, allowing for flexible creation of inspection applications to suit a wide variety of needs.

Intuitive programming
Task View

Easy-to-see
Image View



The screenshot displays the VS Creator software interface. On the left is the 'Task View' containing a 'Main Palette' with categories like Capture, Positioning, Inspection, Communication, and Options. The central 'Image View' shows a simulated circuit board with a highlighted component. On the right is the 'Properties View' with tabs for General, Region, Image Enhance, Detection, Judgment, and Display. The 'Image Enhance' tab is active, showing a histogram and threshold settings. The 'Results' panel at the bottom right shows a table of inspection results.

Judgment	Item Name	Measured Value
True	Area	91308

Easy copy & paste
Shortcut keys available

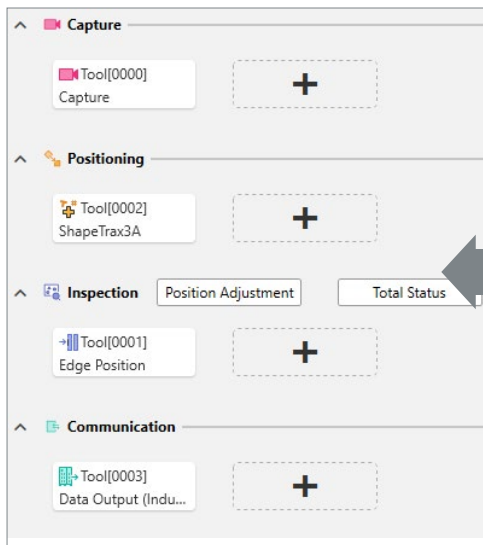
Efficient setup
Properties View



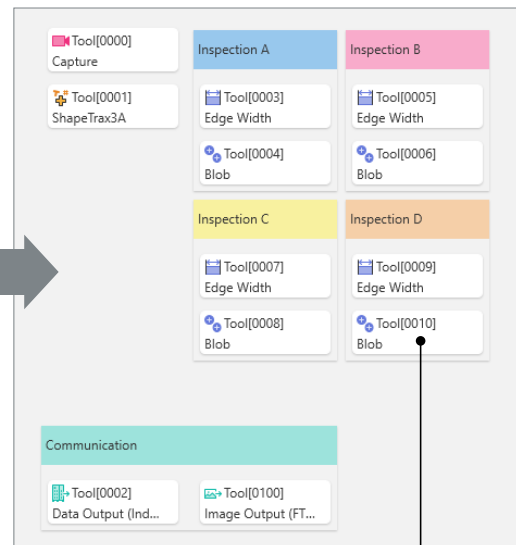
Capable, flexible, and easy to understand

The number of inspection tools needed tends to increase with application complexity. The VS Series keeps it simple and organised.

Step layout made in four steps



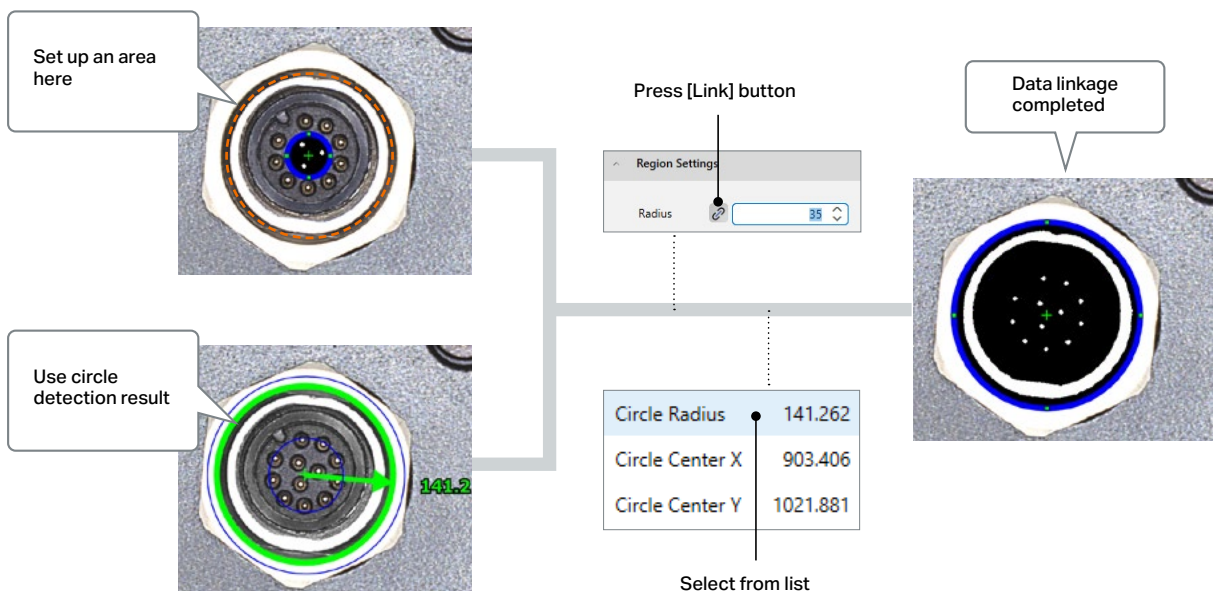
Free layout for easy organisation



Check inspection items at a glance

Link tool results with ease

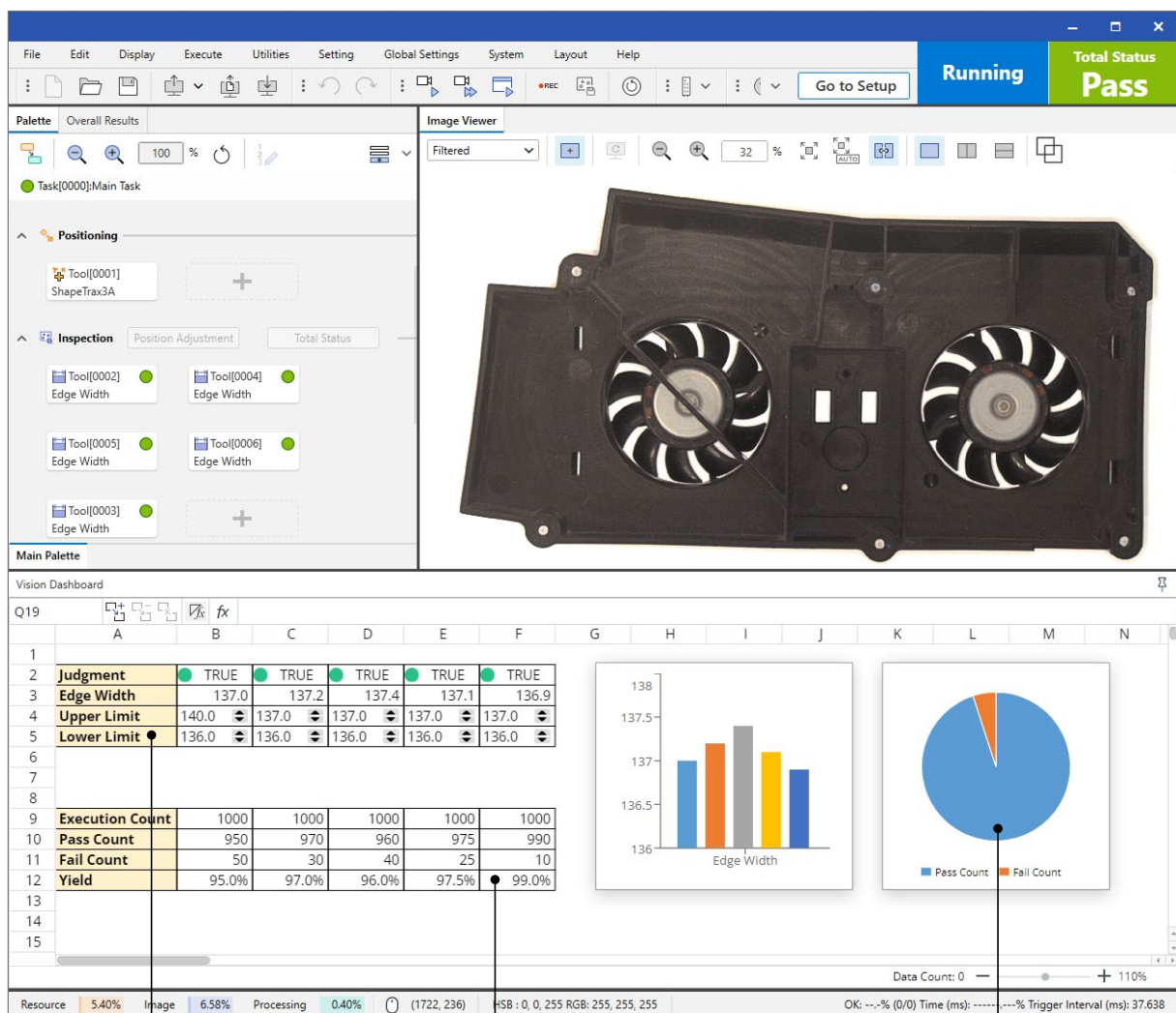
Create complex inspections by linking tools using conditional formatting.



Vision Dashboard — Immediate data utilisation

Instantaneous data utilisation and screen customisation

The Vision Dashboard significantly reduces the time required for data utilisation compared to conventional systems. The various tools and dashboard components also make operation screen customisation incredibly easy.



Lists
to adjust settings

Table format
for yield rate management

Graphs
for quality analysis

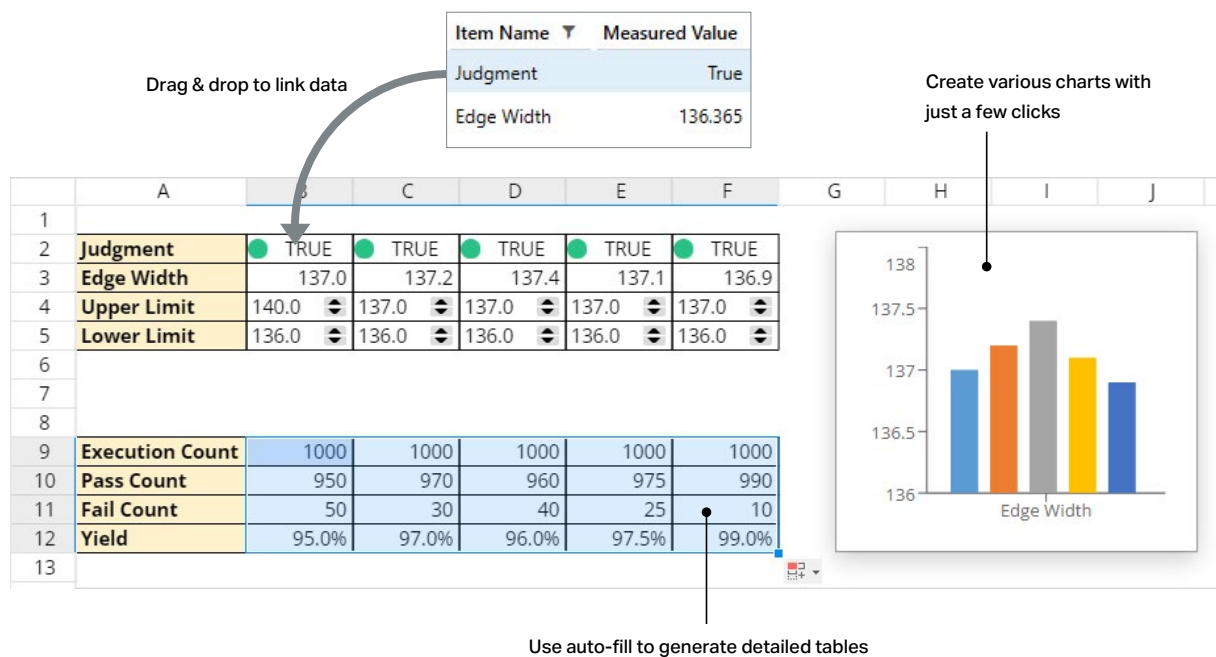


See the video
for more
information



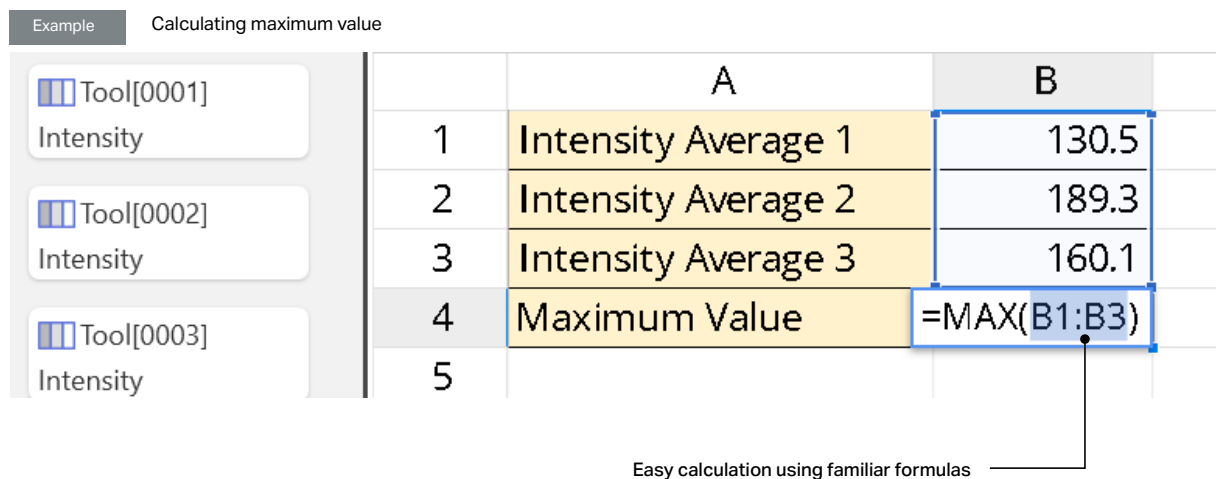
Quickly access necessary data

Create lists and graphs in an instant by utilising standard spreadsheet functions.



Quick calculations using selected data

Formulas can be easily applied using the same rules as common spreadsheet software. You can also calculate multiple cells in a batch.



Intuitive operation screens for onsite use

Easy operation screen customisation

Custom Screen

Dedicated screens that are easy to use for onsite personnel can improve overall efficiency, but making the screens can take a long time. Custom Screen makes it easy to create intuitive, user-friendly operation screens quickly.



Multiple image processing screens displayed side by side

Customisable graphical component locations

History and NG image displays

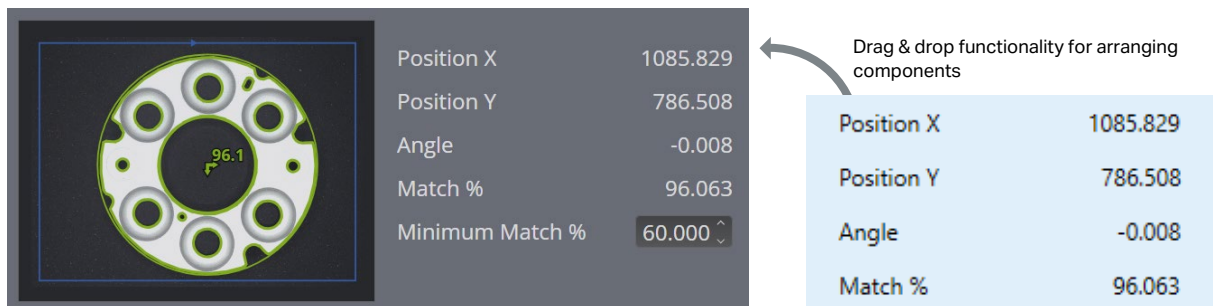
Easy-to-view screens for onsite use

When it comes to factory automation, screens that are clear and easy to read are important. Custom Screen makes it easy to create highly visible components such as buttons, text, and graphs.



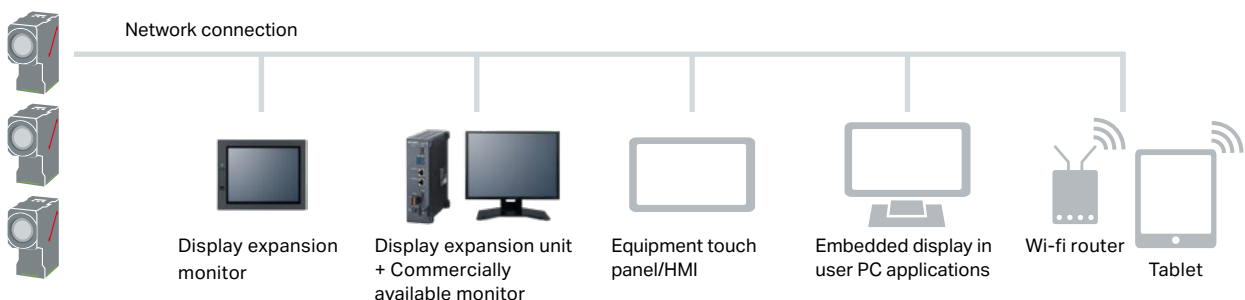
Intuitive screen editing

Values and input boxes can be easily arranged by dragging & dropping or by right-clicking and choosing the desired component. Intuitive PC operations make it easy to bring ideas to life on the screen.



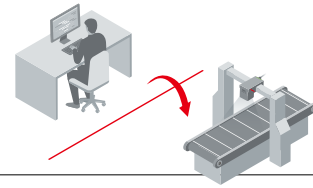
Web-based HMI

Custom operation screens can be used on a wide variety of network-connected display devices. Multiple camera screens can also be displayed on the same device, allowing for a wider choice of usable displays to fit a wider variety of equipment.

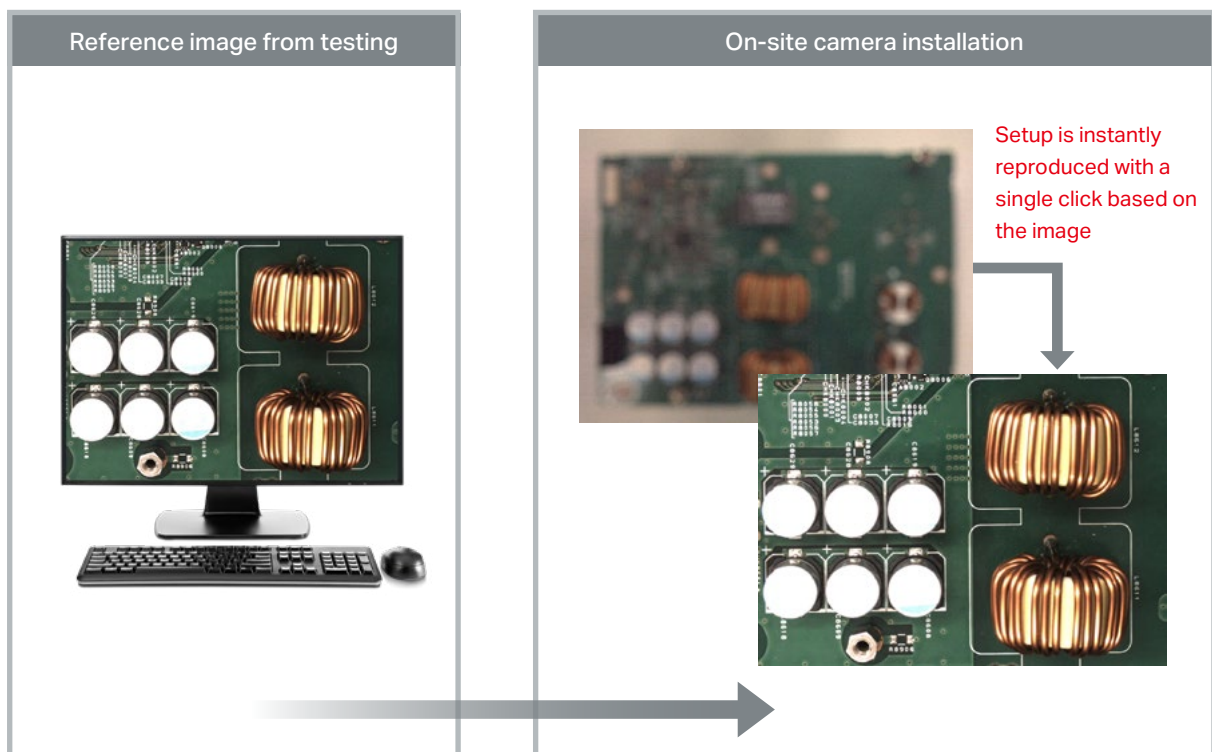


Utilities to facilitate troubleshooting and repeat deployments

Quick startup, scaling, and troubleshooting Capture Replication function



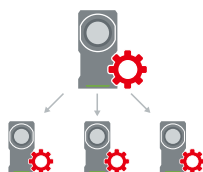
Suppose you want to reproduce lab conditions or need to fix a deployed camera that has been bumped or moved... ZoomTrax's Capture Replication function can instantly replicate the lens setup from a previous test or deployment based on a registered image. This can further reduce the time required to complete installation on-site.



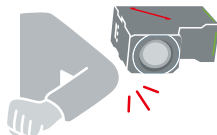
AI reproduces the imaging used in testing by automatically adjusting the field of view, focus, and exposure time.

When setup reproduction is necessary

Reproducing a setup for repeated expansion of lines



Correcting device position when it shifts due to collision



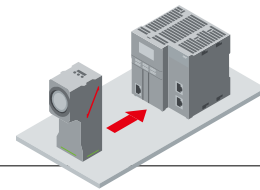
Replicating setup for repair or replacement





Visualising communication

Communication debugging function



Connecting and coordinating communication is a tedious task when setting up a system. The VS Series offers two visualisation utilities that solve common communication issues.

Communication monitor

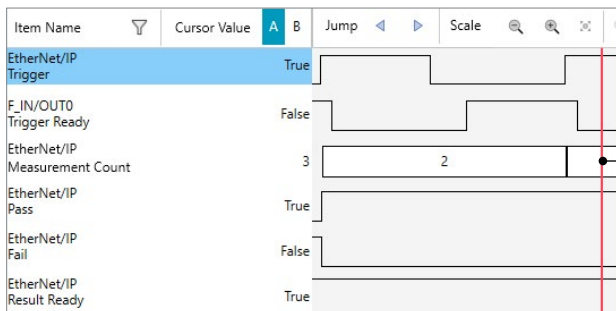
Camera Receive Data			Camera Send Data		
Address	Value	Description	Address	Value	Description
2.3	☐	Result Ack	2.0	☒	Pass
2.4	☒	Go to Run	2.1	☐	Fail
2.5	☐	Go to Setting	2.2	☐	Reserved
2.6	☐	Error0 Clear	2.3	☐	Result Ready
2.7	☐	Error1 Clear	2.4	☐	Run Status
3.0	☐	Command Request	2.5	☐	Reserved

Communication Status: **Connected**

Signal address and status are listed

Simple list format to establish connection with EtherNet/IP™, PROFINET, and PLC Link

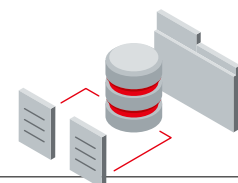
Timing chart monitor



Changes in signals over time can be visualised
Timing errors and missing signals can be analysed

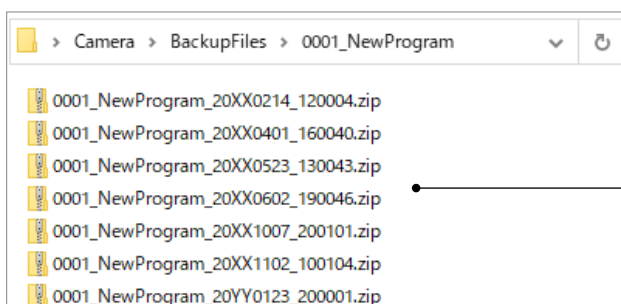
Restore data after accidental overwrite

Automatic backup of setup



The most recent editing history is automatically backed up on the computer. Settings can be restored immediately if a problem arises.

This helps prevent the added work of recreating the settings.



The last 20 automatic backups are stored on the computer.
Settings can be quickly restored from the history.

Three types of smart cameras to support any application

Semi-automated
machines

Versatile standard zoom type for varying fields of view and mounting positions



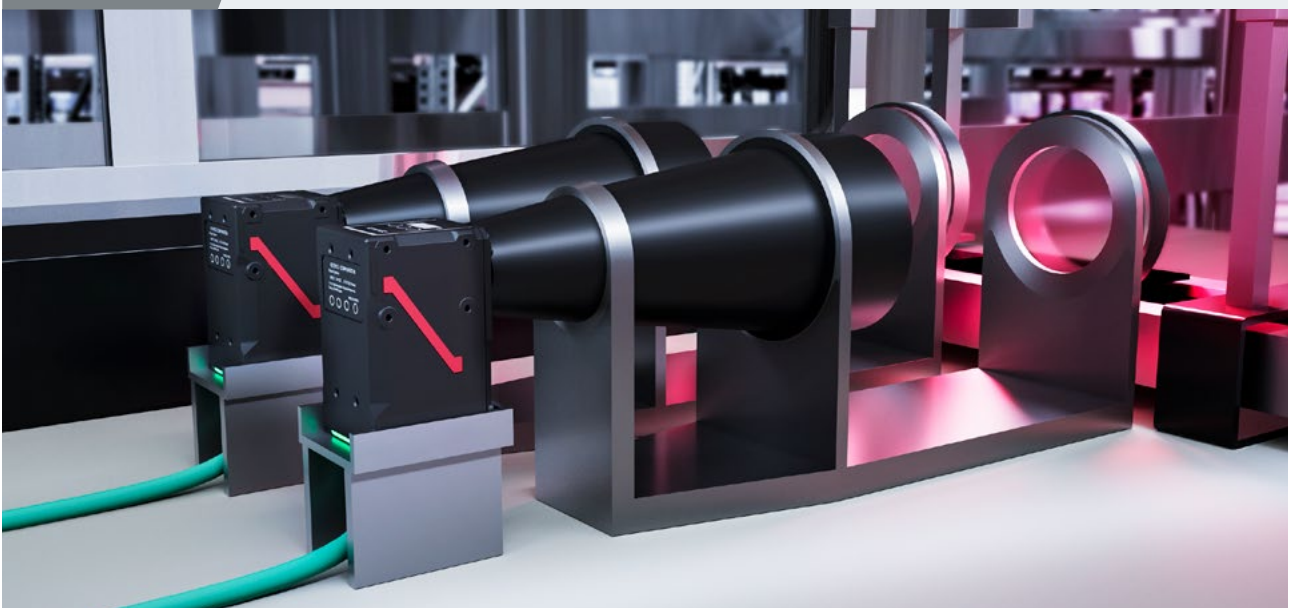
Fully automated
machines

Short-range zoom type for short-range or space-
saving installation needs



Customisation for
specialised
purposes

C-mount type when specialty lenses are necessary



Easy four-step process for selecting your optimal hardware

STEP.1

Select the camera type

Zoom smart camera

This unit supports a wide range of applications and field-of-view sizes.

The zoom function saves you from having to make difficult optical adjustments.

Standard zoom
type
VS-L



Short-range zoom
type
VS-S



C-mount smart camera

Use a C-mount smart camera if you want to select from a variety of lenses for different field-of-view sizes and installation distances.

C-mount type
VS-C



Various lenses



STEP.2

Select the camera resolution

Selecting a zoom camera

Select the resolution according to your application.

1.6 MP

Simple applications
such as presence
check

3.2 MP

Assembly inspection,
character recognition,
etc.

5 MP

For inspections that
require high accuracy,
such as measurement
and appearance
inspections

15 MP

Ultra-high-precision
inspections
or for large targets

Supported target sizes and resolutions are listed on the field of view chart on page 30.



Access the Item
Selector



STEP.3

Select the lighting

Smart ring lighting, High intensity

The lineup of camera-integrated lighting covers a wide range of applications.



Smart ring lighting,
High intensity
CA-DEW10X (white)



Smart ring lighting,
High intensity
CA-DER10X (red)



Smart ring lighting,
High intensity
CA-DEB10X (blue)



Smart ring lighting,
High intensity
CA-DEIR10X
(near-infrared)

Additional Lighting Options

The lineup includes lighting of various shapes and sizes, which can be matched to your application and detection requirements.



Multi-spectrum
lighting
CA-DRMxX



Multi-angle lights
(ring/square)
CA-DRxM
CA-DQxM



Bar lights
CA-DB
CA-DBxW



Back lights
CA-DS



Coaxial lights
(on-axis)
CA-DX



Dome lights
CA-DD



Spot lights
CA-DP

STEP.4

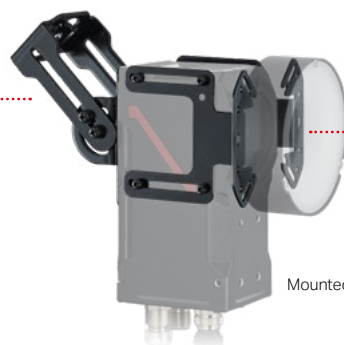
Select optional mounting

Different mounting jigs to meet your specific installation needs

* Other mounting jigs are listed on page 31.



Mounting bracket, Angle
adjustable, for high
performance camera
OP-88814



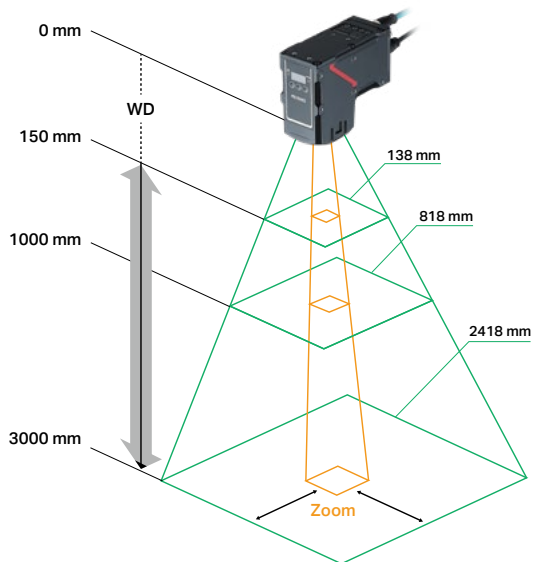
Mounted view



Mounting adapter for lighting,
for high performance zoom
camera
OP-88816

Field of view information

Field of view information for standard zoom smart cameras



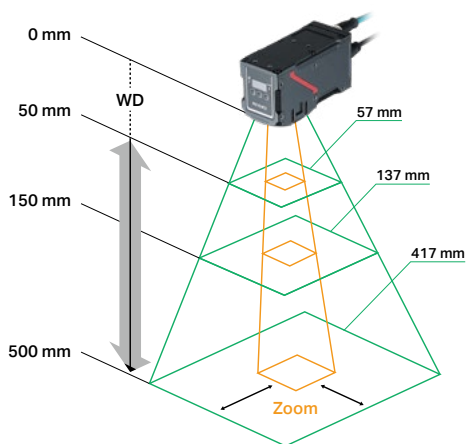
	Working distance	Field of view	Pixel resolution
VS-L160	150 mm	18 to 138 mm	12 to 95 μ m
	1000 mm	94 to 818 mm	65 to 568 μ m
	3000 mm	274 to 2418 mm	190 to 1679 μ m

VS-L320	150 mm	25 to 138 mm	12 to 67 μ m
	1000 mm	134 to 818 mm	65 to 399 μ m
	3000 mm	390 to 2418 mm	190 to 1180 μ m

VS-L500	150 mm	31 to 138 mm	12 to 56 μ m
	1000 mm	162 to 818 mm	65 to 331 μ m
	3000 mm	470 to 2418 mm	190 to 981 μ m

VS-L1500	150 mm	55 to 138 mm	13 to 31 μ m
	1000 mm	289 to 818 mm	65 to 185 μ m
	3000 mm	839 to 2418 mm	190 to 549 μ m

Field of view information for short-range zoom smart cameras



	Working distance	Field of view	Pixel resolution
VS-S160	50 mm	16 to 57 mm	11 to 39 μ m
	150 mm	37 to 137 mm	25 to 95 μ m
	500 mm	116 to 417 mm	80 to 289 μ m

VS-S320	50 mm	22 to 57 mm	11 to 28 μ m
	150 mm	54 to 137 mm	26 to 67 μ m
	500 mm	166 to 417 mm	81 to 203 μ m

VS-S500	50 mm	27 to 57 mm	11 to 23 μ m
	150 mm	66 to 137 mm	26 to 55 μ m
	500 mm	200 to 417 mm	81 to 169 μ m

VS-S1500	50 mm	49 to 57 mm	11 to 13 μ m
	150 mm	118 to 137 mm	26 to 31 μ m
	500 mm	358 to 417 mm	81 to 94 μ m

Supported communication protocol(s)
Communication networks (standard equipment)

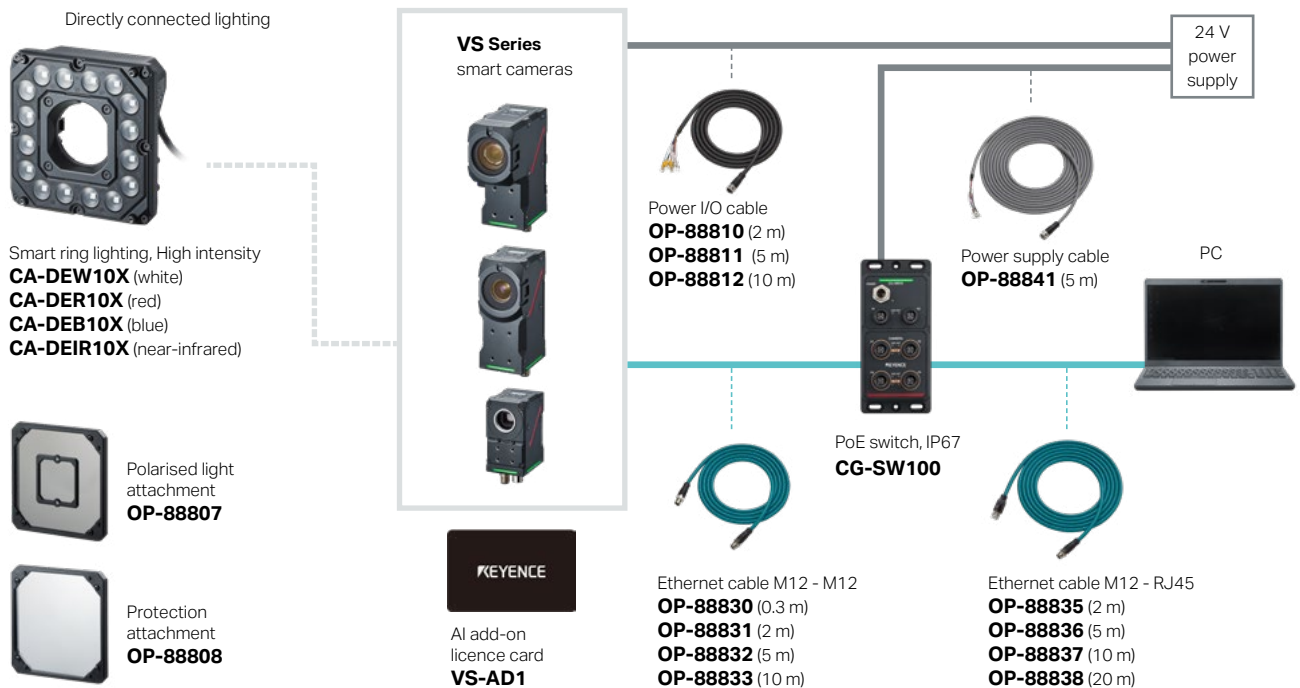
PLC LINK

EtherNet/IP

PROFINET

OPC UA

TCP/IP



Lighting accessories



Refer to the Vision System Peripheral Equipment Lighting Lineup catalogue for more information on the lighting lineup.

Camera mounting accessories



Lens accessories



Refer to the Vision System Peripheral Equipment Lineup catalogue for more information on the lens lineup.

Display accessories



Other accessories



Specifications

Standard zoom smart camera/Short-range zoom smart camera

Model		VS-LxxxMX				VS-LxxxCX				VS-SxxxMX				VS-SxxxCX				
		160	320	500	1500	160	320	500	1500	160	320	500	1500	160	320	500	1500	
Storage	Internal storage	3 GB	7 GB	7 GB	7 GB	3 GB	7 GB	7 GB	7 GB	3 GB	7 GB	7 GB	7 GB	3 GB	7 GB	7 GB	7 GB	
	microSD card (external)*1	Max. 64 GB																
Image sensor	Frame rate	85 fps	83 fps	81 fps	48 fps	85 fps	83 fps	81 fps	48 fps	85 fps	83 fps	81 fps	48 fps	85 fps	83 fps	81 fps	48 fps	
Lens	Mount	Lens-integrated type																
	Max. optical magnification	× 8.75	× 6.0	× 5.0	× 2.75	× 8.75	× 6.0	× 5.0	× 2.75	× 3.5	× 2.4	× 2.0	× 1.1	× 3.5	× 2.4	× 2.0	× 1.1	
	Installation distance*2	150 mm to 5000 mm								50 mm to 500 mm								
I/O connection	Number of inputs	1 (insulated)																
	Number of outputs	3 (insulated)																
	Number of inputs/outputs	2 (insulated) (input and output switched via settings)																
I/F	Ethernet	PoE IEEE 802.3at, M12 X cable × 1																
	I/O	1 In/3 Out/2 In, Out M12 A cable × 1																
	Lighting interface	Dedicated lighting connection interface × 1																
	External media	microSD card slot × 1 (UHS-1)																
Functions	Imaging mode	1.6 M	1.6 M 3.2 M	1.6 M 3.2 M 5 M	1.6 M 3.2 M 5 M 15 M	1.6 M	1.6 M 3.2 M	1.6 M 3.2 M 5 M	1.6 M 3.2 M 5 M 15 M	1.6 M	1.6 M 3.2 M	1.6 M 3.2 M 5 M	1.6 M 3.2 M 5 M 15 M	1.6 M	1.6 M 3.2 M	1.6 M 3.2 M 5 M	1.6 M 3.2 M 5 M 15 M	
	Output image pixels	(1.6 M mode) 1440×1072 1072×1440 1248×1248 1664×928 928×1664 2160×720 720×2160	(3.2 M mode) 2048×1536 1536×2048 1776×1776 2368×1328 1328×2368 3072×1024 1024×3072	(5 M mode) 2544×1904 1904×2544 2224×2224 2976×1664 1664×2976 3824×1280 1280×3824	(15 M mode) 4400×3296 1248×1248 1664×928 928×1664 2160×720 720×2160	(1.6 M mode) 1440×1072 1072×1440 1248×1248 1664×928 928×1664 2160×720 720×2160	(3.2 M mode) 2048×1536 1536×2048 1776×1776 2368×1328 1328×2368 3072×1024 1024×3072	(5 M mode) 2544×1904 1904×2544 2224×2224 2976×1664 1664×2976 3824×1280 1280×3824	(15 M mode) 4400×3296 1248×1248 1664×928 928×1664 2160×720 720×2160	(1.6 M mode) 1440×1072 1072×1440 1248×1248 1664×928 928×1664 2160×720 720×2160	(3.2 M mode) 2048×1536 1536×2048 1776×1776 2368×1328 1328×2368 3072×1024 1024×3072	(5 M mode) 2544×1904 1904×2544 2224×2224 2976×1664 1664×2976 3824×1280 1280×3824	(15 M mode) 4400×3296 1248×1248 1664×928 928×1664 2160×720 720×2160	(1.6 M mode) 1440×1072 1072×1440 1248×1248 1664×928 928×1664 2160×720 720×2160	(3.2 M mode) 2048×1536 1536×2048 1776×1776 2368×1328 1328×2368 3072×1024 1024×3072	(5 M mode) 2544×1904 1904×2544 2224×2224 2976×1664 1664×2976 3824×1280 1280×3824	(15 M mode) 4400×3296 1248×1248 1664×928 928×1664 2160×720 720×2160	
	Exposure time	0.037 msec to 1000 msec																
	Image correction functions	Gain, Offset, Gamma correction, White balance, Fine HDR																
	Indicators	OLED display, Status LED, Ethernet LED (LINK/ACT), SD card access indicator																
	Buttons	Operation buttons × 3																
	Installation support functions	Pointer (Class 1 laser product*3), Angle monitor																
	Power supply	Voltage	24 V +25%/−20% or PoE (IEEE802.3at Power Class 4)															
		Current consumption (camera only)	0.97 A, 18.7 W (for 19.2 V)/0.78 A, 18.7 W (for 24 V)															
		Current consumption (including lighting)	4.7 A, 89.7 W (for 19.2 V)/3.8 A, 89.7 W (for 24 V)															
Current consumption (With CA-DEx10X connected)*4		11.3 A, 216.7 W (for 19.2 V)/9.1 A, 216.7 W (for 24 V)																
Other	Weight	Approx. 700 g								Approx. 570 g								
	Size (H×W×D)	122.3 mm × 52.6 mm × 99.1 mm								122.3 mm × 52.6 mm × 69.2 mm								
	Enclosure rating	IP67 (IEC60529)																
	Materials*5	Case: Aluminium die-casting, Front cover: Glass																
	Case temperature*6	0°C to 65°C																
	Operating ambient humidity	85% RH or below (no condensation)																
	Standard certifications	CE, FCC, NRTL, KC, UKCA																
Vibration resistance	10 to 500 Hz; Power spectral density: 0.05 G2/Hz; X, Y, and Z directions, 0.5 hours (IEC60068-2-64)																	
Shock resistance	50 G, 3 times in each of the 6 directions (IEC60068-2-27)																	

*1 Using the configuration software, format the media before use. *2 VS-LxxxMX/CX's field-of-view measuring range is 150 mm to 3000 mm. VS-SxxxMX/CX's field-of-view measuring range is 50 mm to 500 mm. *3 FDA (CDRH) Part 1040.10 (The laser classification is implemented based on IEC 60825-1 in accordance with the requirements of Laser Notice, IEC60825-1. *4 Maximum instantaneous current. Average current is below current consumption (including lighting).
 *5 ESD-Safe, IEC61340-5-1 compliant. *6 Reference - Case temperature 65°C with 200 mm × 200 mm × 10 mm aluminium board and at ambient temperature of 40°C.
 • The "MX" model name suffix indicates a monochrome camera model, and the "CX" suffix indicates a colour camera model.
 • 1.6-megapixel cameras do not support AI Segmentation or AI Detection Fine Mode.

C-mount smart camera

Model		VS-C160M / CX	VS-C320M / CX	VS-C500M / CX	VS-C1500M / CX	VS-C2500M / CX
Storage	Internal storage	3 GB	7 GB	7 GB	7 GB	7 GB
	microSD card (external)*1	Max. 64 GB				
Image sensor	Image size	1/2.9"	1/1.8"	2/3"	1.1"	1.1"
	Pixel size	3.45 µm	3.45 µm	3.45 µm	2.5 µm	2.5 µm
Lens	Frame rate	83 fps	62 fps	55 fps	40 fps	28 fps
	Mount	C-mount				
I/O connection	Number of inputs	1 (insulated)				
	Number of outputs	3 (insulated)				
I/F	Number of inputs/outputs	2 (insulated) (input and output switched via settings)				
	Ethernet	PoE IEEE 802.3at, M12 X cable × 1				
I/O	I/O	1 In/3 Out/2 In, Out M12 A cable × 1				
	Lighting interface	Dedicated lighting connection interface × 1				
External media	External media	microSD card slot × 1				
Output image pixels	Output image pixels	1440 × 1072	2048 × 1536	2448 × 2048	4400 × 3296	5120 × 5120
	Exposure time	0.015 msec to 1000 msec	0.015 msec to 1000 msec	0.015 msec to 1000 msec	0.008 msec to 1000 msec	0.008 msec to 1000 msec
Image correction functions	Image correction functions	Gain, Offset, Gamma correction, White balance, Fine HDR				
	Indicators	OLED display, Status LED, Ethernet LED (LINK/ACT), SD card access indicator				
Buttons	Buttons	Operation buttons × 3				
	Installation support functions	Angle monitor				
Voltage	Voltage	24 V +25%/−20% or PoE (IEEE802.3at Power Class 4)				
	Current consumption (camera only)	0.97 A, 18.7 W (for 19.2 V)/0.78 A, 18.7 W (for 24 V)				
Current consumption (including lighting)	Current consumption (including lighting)	4.7 A, 89.7 W (for 19.2 V)/3.8 A, 89.7 W (for 24 V)				
	Current consumption (With CA-DEx10X connected)*2	11.3 A, 216.7 W (for 19.2 V)/9.1 A, 216.7 W (for 24 V)				
Weight	Weight	Approx. 420 g				
	Size (H×W×D)	93.2 mm × 52.6 mm × 66 mm				
Enclosure rating	Enclosure rating	IP67 (IEC60529)				
	Materials*3	Case: Aluminium die-casting, Front cover: Glass				
Case temperature*4	Case temperature*4	0°C to 65°C				
	Operating ambient humidity	85% RH or below (no condensation)				
Standard certifications	Standard certifications	CE, FCC, NRTL, KC, UKCA				
	Vibration resistance	10 to 500 Hz; Power spectral density: 0.05 G2/Hz; X, Y, and Z directions, 0.5 hours (IEC60068-2-64)				
Shock resistance	Shock resistance	50 G, 3 times in each of the 6 directions (IEC60068-2-27)				

*1 Using the configuration software, format the media before use. *2 Maximum instantaneous current. Average current is below current consumption (including lighting). *3 ESD-Safe, IEC61340-5-1 compliant.
 *4 Reference - Case temperature 65°C with 200 mm × 200 mm × 10 mm aluminium board and at ambient temperature of 40°C.
 • The "MX" model name suffix indicates a monochrome camera model, and the "CX" suffix indicates a colour camera model.
 • 1.6-megapixel cameras do not support AI Segmentation or AI Detection Fine Mode.

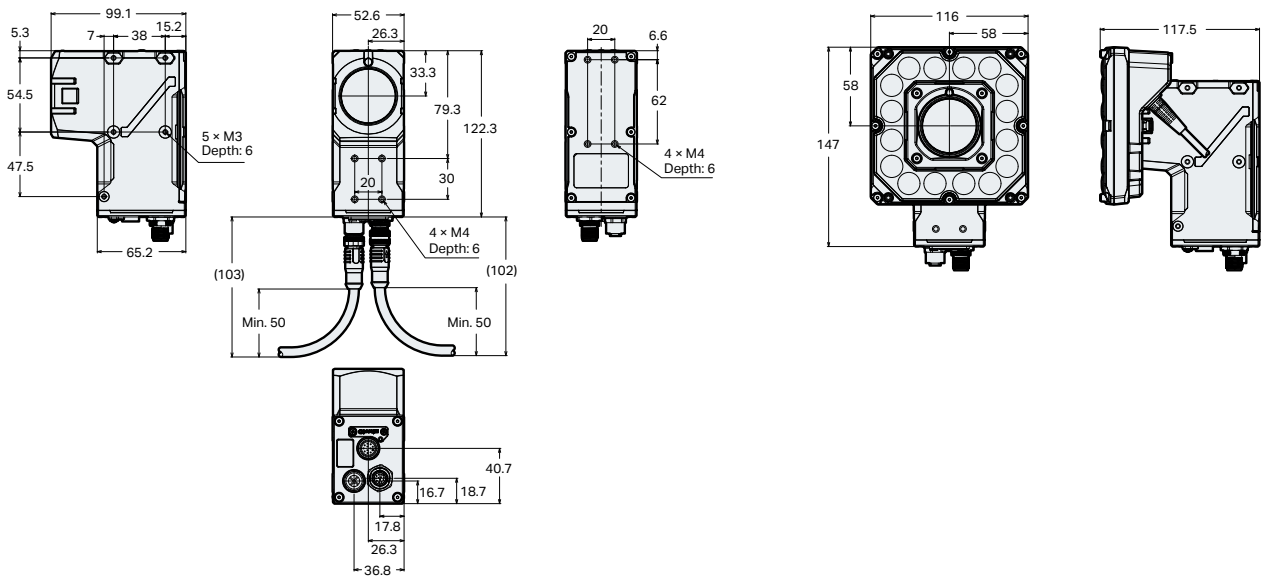
Standard zoom smart camera

Camera

VS-L1500MX / L1500CX / L500MX / L500CX / L320MX / L320CX / L160MX / L160CX

Smart ring lighting, High intensity

When used with **CA-DEW10X/DER10X/DEB10X/DEIR10X**



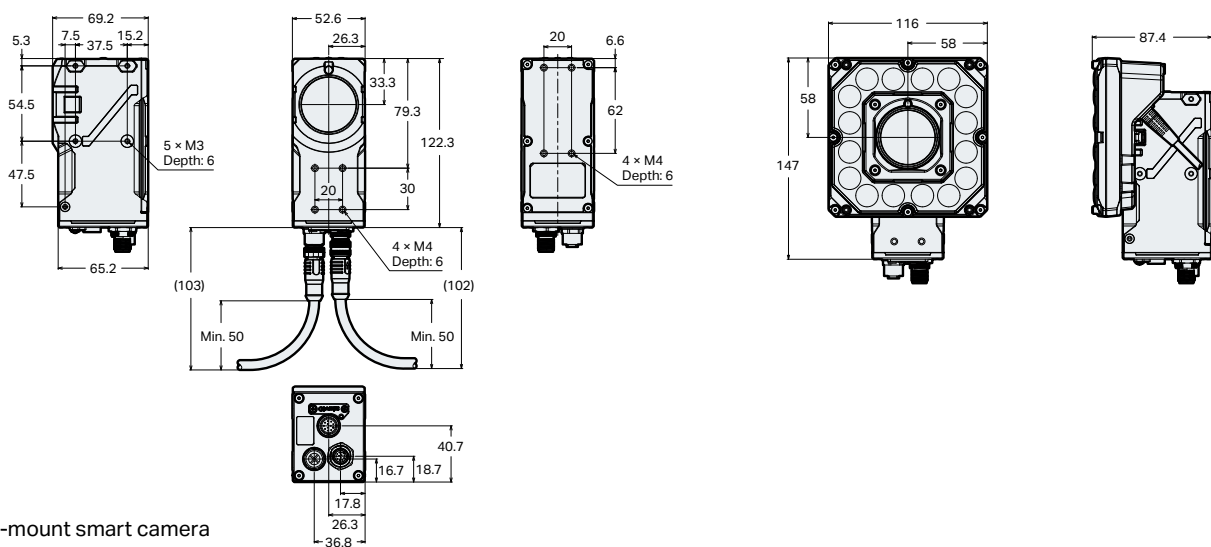
Short-range zoom smart camera

Camera

VS-S1500MX / S1500CX / S500MX / S500CX / S320MX / S320CX / S160MX / S160CX

Smart ring lighting, High intensity

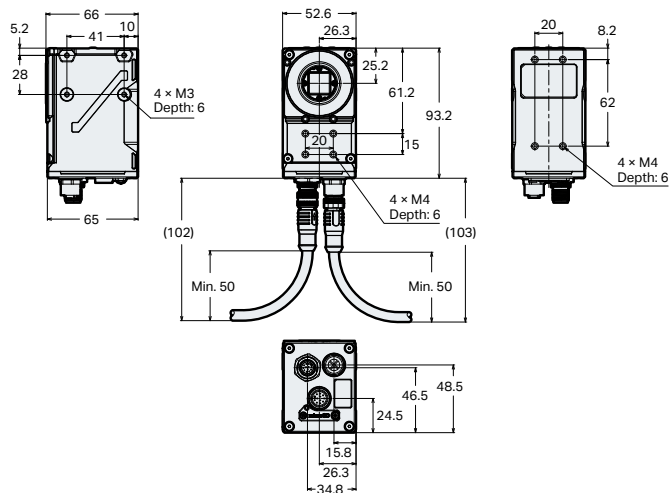
When used with **CA-DEW10X/DER10X/DEB10X/DEIR10X**



C-mount smart camera

Camera

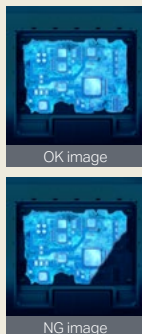
VS-C2500MX / C2500CX / C1500MX / C1500CX / C500MX / C500CX / C320MX / C320CX / C160MX / C160CX



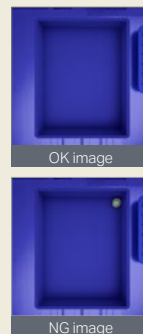
Automobiles/Metals

Presence check

■ Inspection for desiccant on ECU

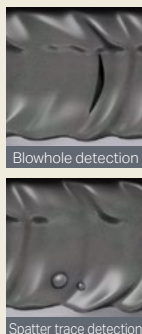


■ Remaining target detection

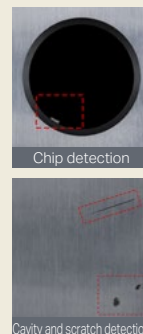
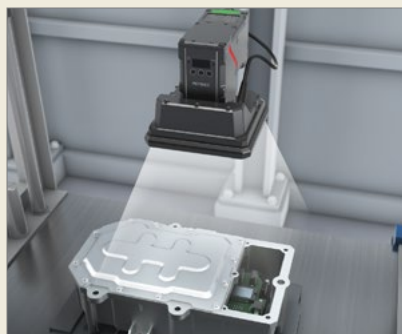


Appearance inspection

■ Laser welding appearance inspection

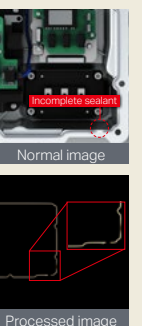


■ Die-cast product appearance inspection



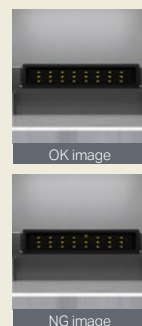
Appearance inspection

■ Sealant application inspection



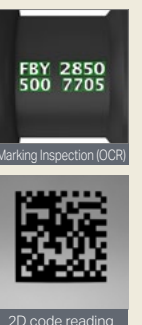
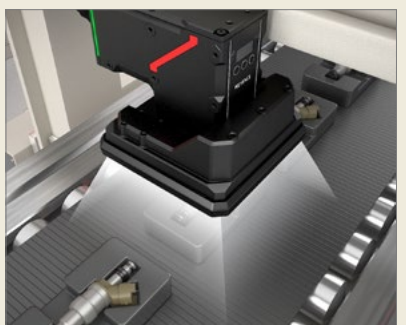
Dimension inspection

■ Connector terminal bend inspection



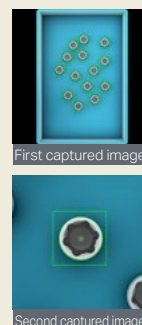
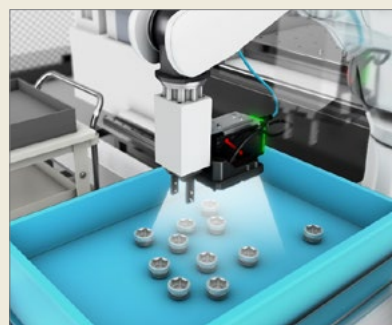
Identification inspection

■ Injector part number OCR and 2D code reading



Vision-guided robotics

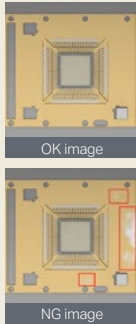
■ Two-step image-based picking of small parts on pallets



Electronic components

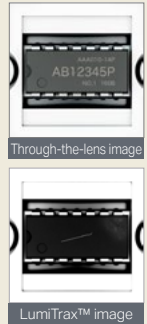
Presence check

Lead frame plating inspection



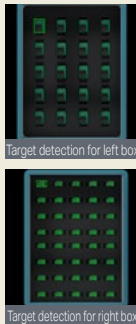
Appearance inspection

IC mould defect/blister inspection



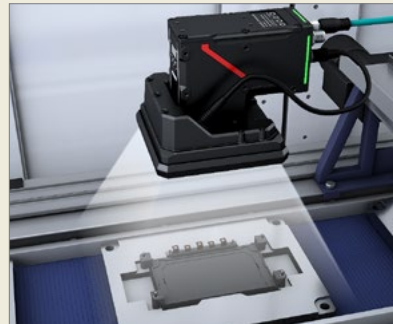
Vision-guided robotics

Stack picking of multiple targets



Appearance inspection

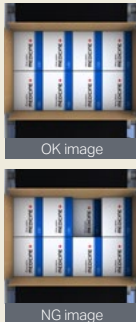
Bus bar appearance inspection



Food / Pharmaceuticals / Commodities

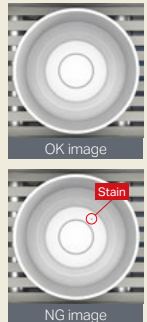
Presence check

Commodity alignment inspection (missing/incorrectly positioned targets)



Appearance inspection

Food tray appearance inspection



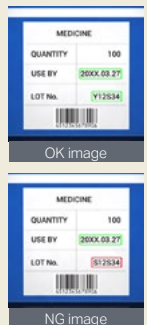
Appearance inspection

Packing seal width inspection



Character recognition

Medicine text and barcode inspection



A Game-Changing Vision System



KEYENCE CORPORATION

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